Unified Modeling Language (UML) Structural & Behavioral Models

Academic Grievance, No-Dues & Clearance Automation Engine

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OMG UML 2.5 Modeling Rules & Standards Compliance Matrix

This document conforms strictly to the **Object Management Group (OMG) UML 2.5 Superstructure Specification**:

Diagram Type	Diagram Classification	Formal OMG UML 2.5 Rules & Semantics Applied	Section
Use Case Diagram	Behavioral	Actors outside system boundary; use cases within boundary; precise `<<include>>` (mandatory functional dependency) and `<<extend>>` (conditional extension) stereotypes.	Section 1
Class Diagram	Structural	3-compartment notation; member visibility indicators (`+` Public, `#` Protected, `-` Private); precise relationship semantics: Generalization (`<	`--`), `Realization (`..
Activity Diagram	Behavioral	Concurrency fork and join bars; decision and merge diamonds; action states; initial/final activity nodes; swimlane encapsulation of gate authorities.	Section 3
Sequence Diagram	Interaction	Participant lifelines; strictly descending chronological time axis; synchronous calls (`-->>`), returns (`-->>`); interaction use fragments (`alt` , `opt` , `loop`).	Section 4
State Machine Diagram	Behavioral	State transitions triggered by events and guards (`Event [guard] / action`); composite states (`Gate 1` , `Gate 2` , `Gate 3`); initial pseudo-states (`[*]`).	Section 5
Component & Deployment	Structural / Physical	3-tier boundary separation (Client UI $\rightarrow$ Application Services $\rightarrow$ Persistent Data Store); interface connectors and protocol definitions.	Section 6

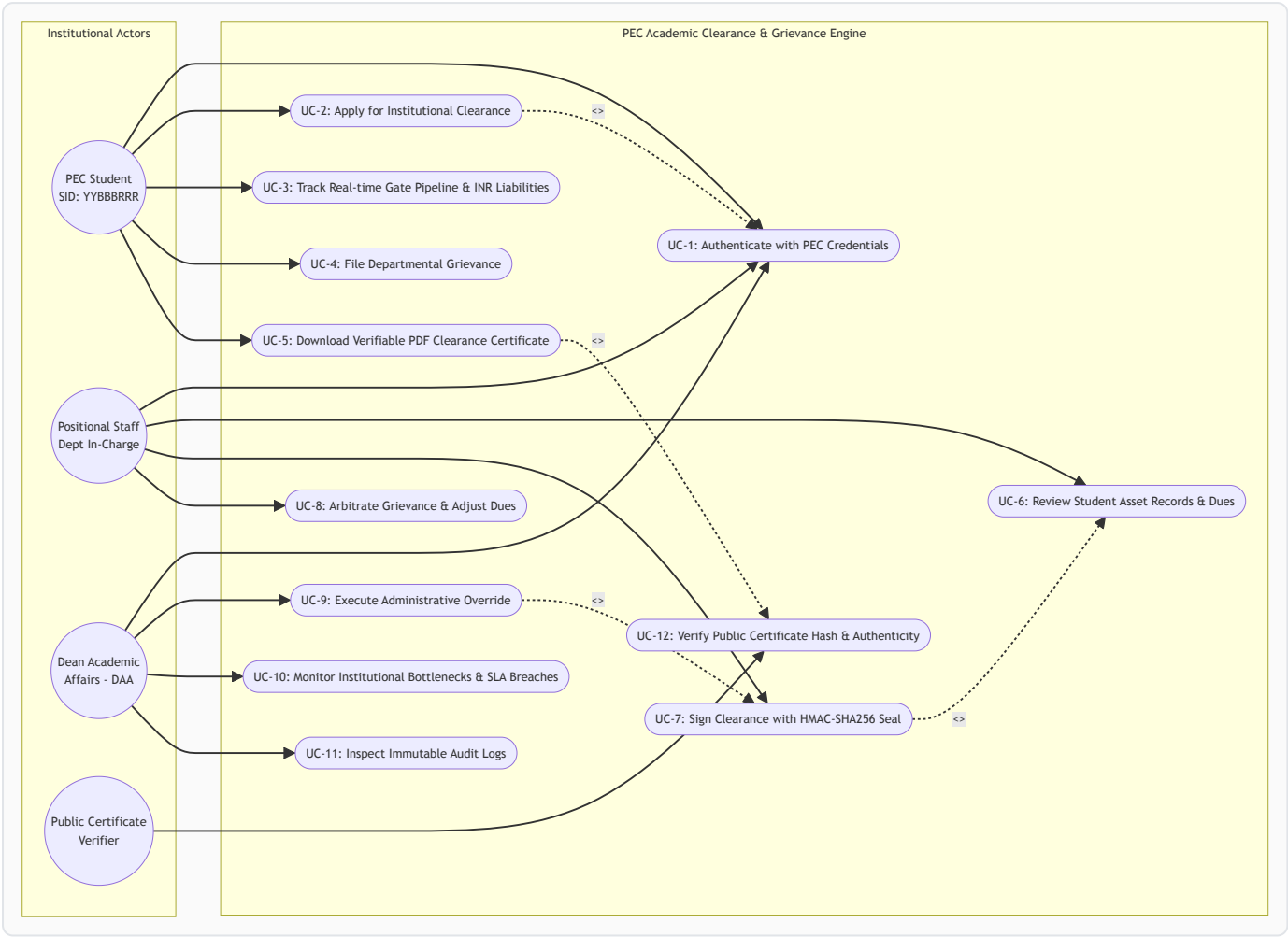
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1. Use Case Model & Diagram

1.1 Actors & Role Boundaries

- **Student Actor:** Registered PEC student identified by an 8-digit SID (e.g., `22103042`).
- **Positional Staff Actor:** Departmental authority operating under permanent institutional designations (*Librarian*, *Chief Warden*, *Gymkhana President*, *HOD Labs*, *Accounts Officer*).
- **Dean Academic Affairs (DAA / Admin Actor):** Executive authority holding override privileges and institutional audit oversight.
- **Public Certificate Verifier Actor:** External unauthenticated party validating clearance credentials.

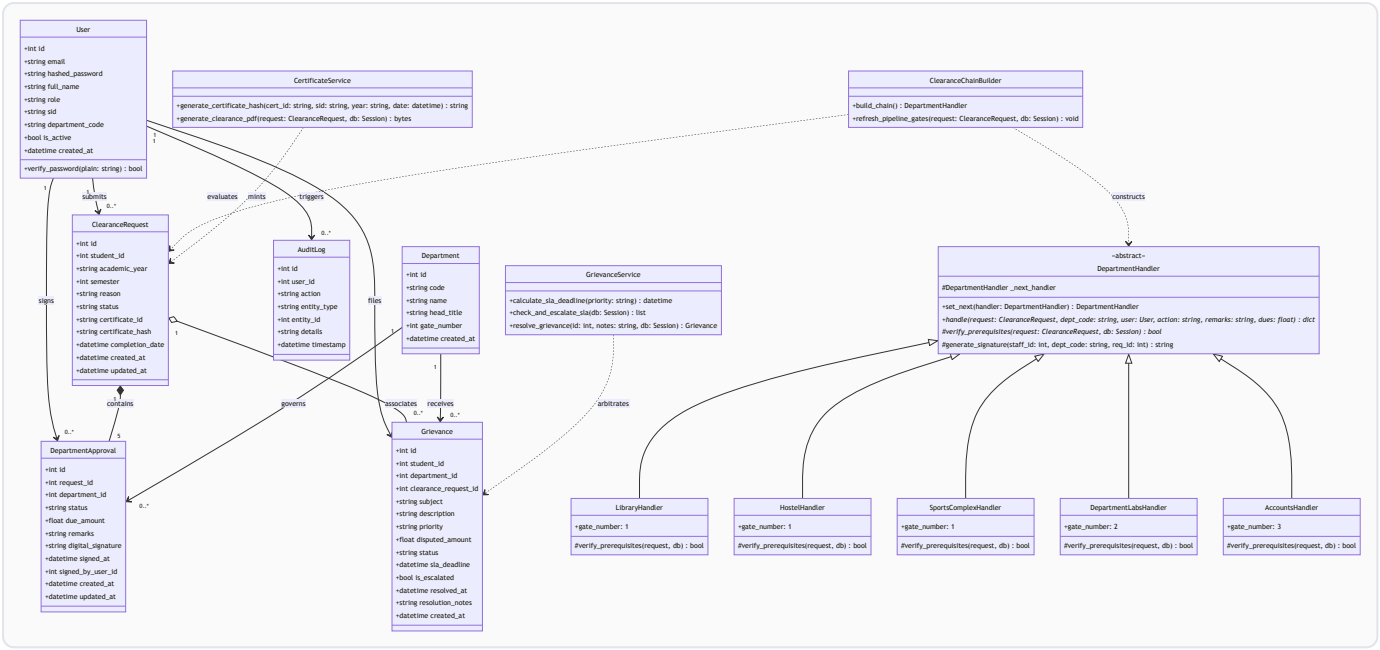
1.2 Use Case Diagram



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2. Static Structural Model: Class Diagram

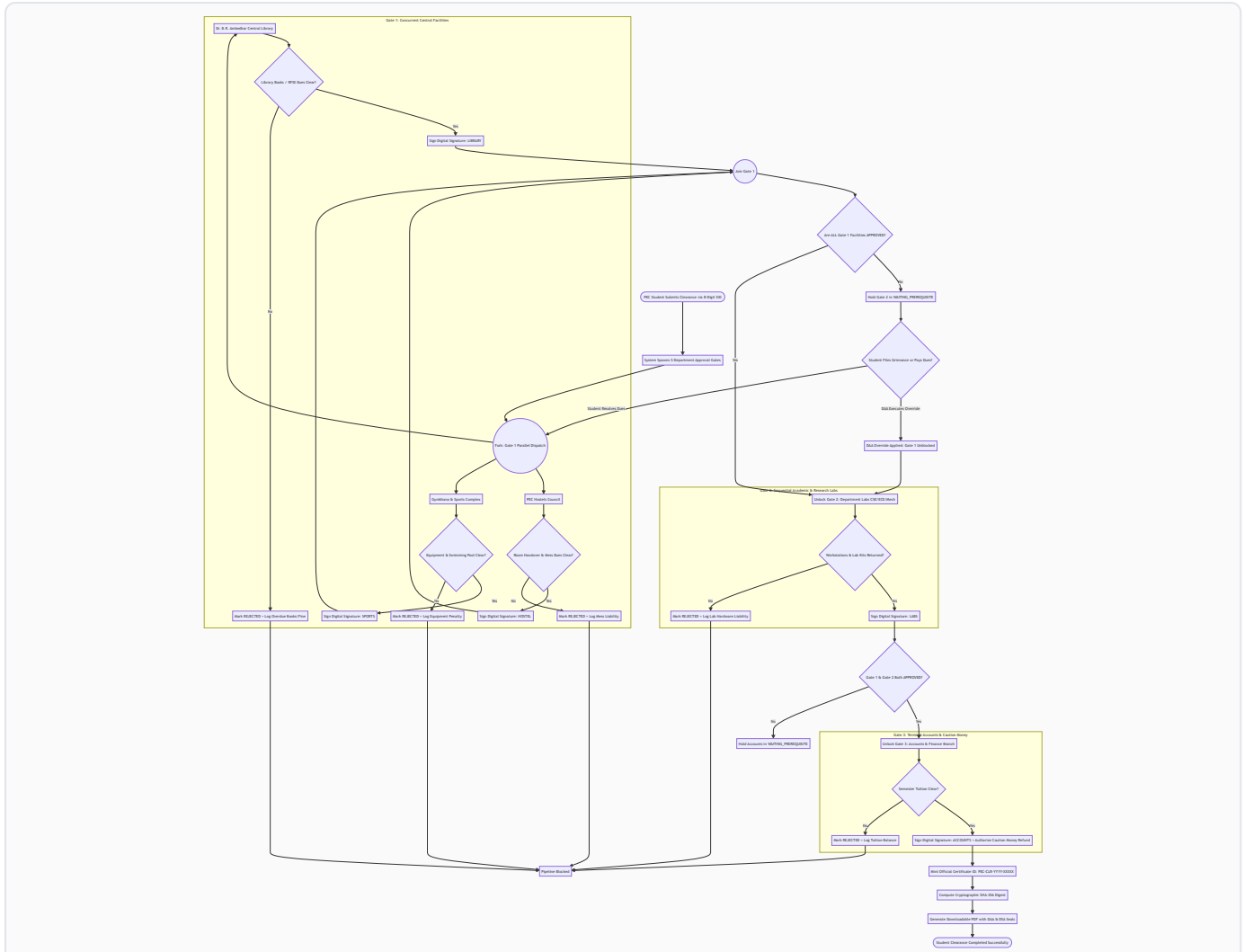
The class diagram captures the object-oriented domain entities, the Chain of Responsibility behavioral pattern handlers, cryptographic helper utilities, and service orchestration layers:



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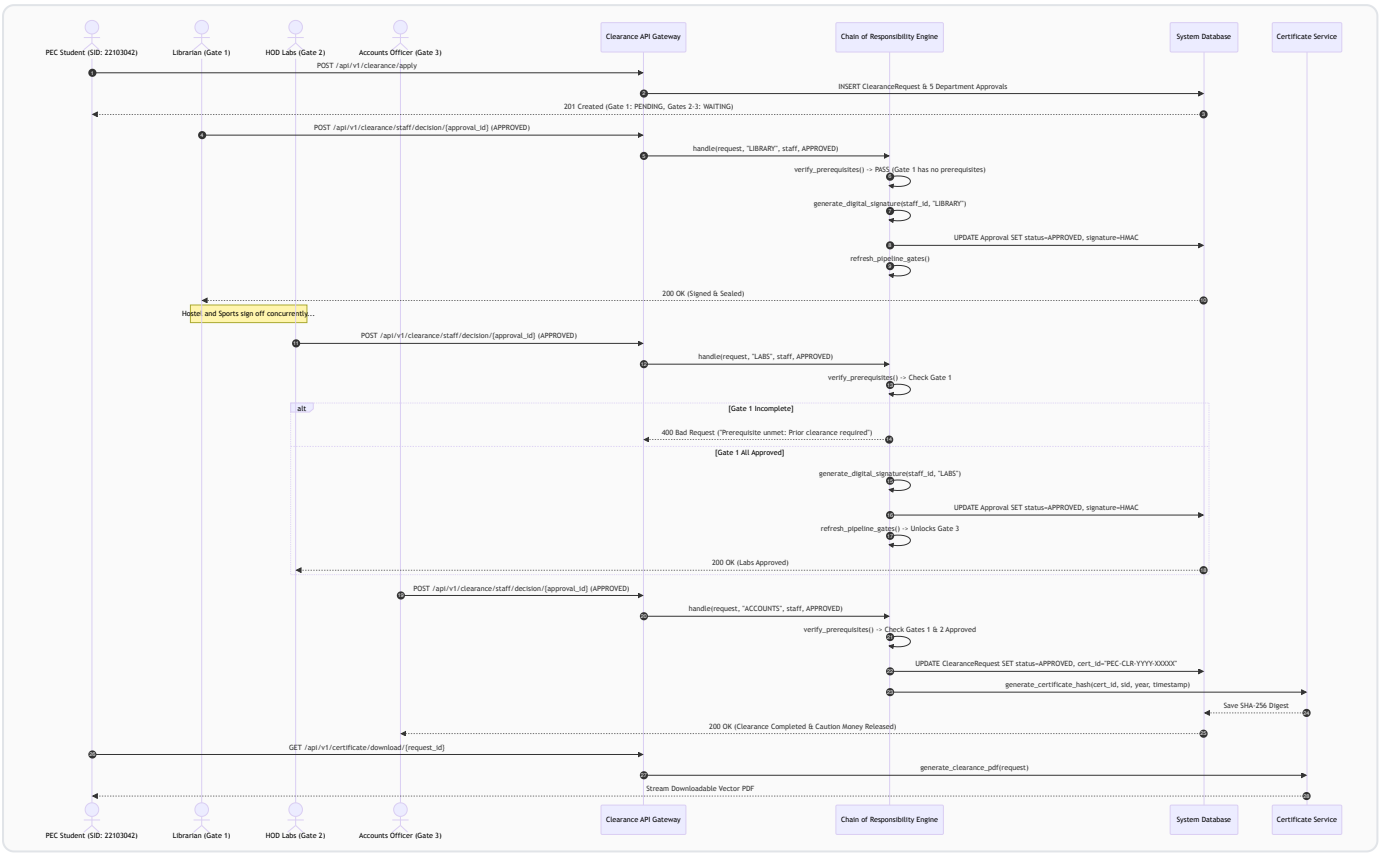
3. Dynamic Behavioral Model: Activity Diagram

This activity diagram captures the end-to-end lifecycle of a PEC clearance request, illustrating parallel concurrency forks, barrier joins, and sequential dependency gates:

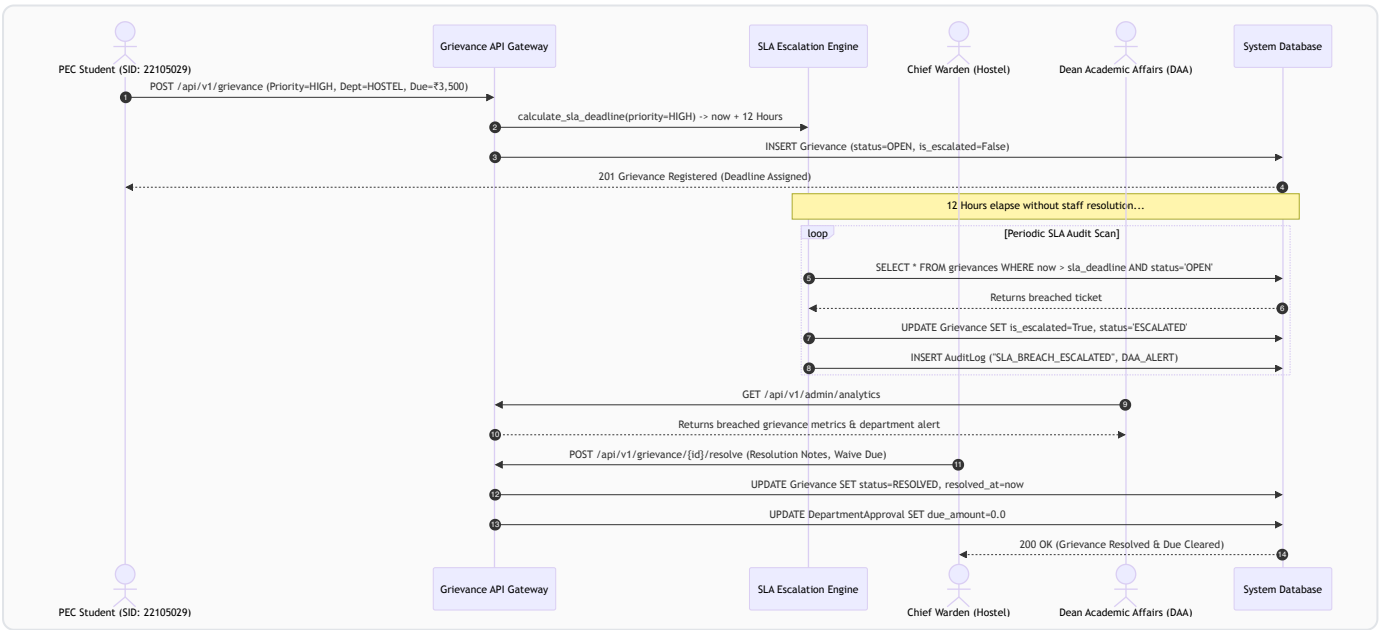

Gate 1: Concurrent Central Facilities
Dr. K.K. Ambedkar Central Library
Library Books / RFID Dues Clear?
Yes
No
Sign Digital Signature: LIBRARY
Join Gate 1
Are ALL Gate 1 Facilities APPROVED?
Yes
No
PEC Student Submits Grievance via 8-Digit SID
Hold Gate 2 in MUTING_PREREQUISITE
System Spawns 5 Departmental Approval Gates
Student Files Grievance or Pays Dues?
Yes
Student Resolves Dues
Bilal Exercises Override
Park: Gate 1 Parallel Dispatch
Bilal Override Approved: Gate 1 Unblocked
Gate 2: Concurrent Academic & Research Labs
Unlock Gate 2: Departmental Labs CSE/ECE/Mech
Gymkhana & Sports Complex
PEC Hostels Council
Workstations & Lab Kits Returned?
Equipment & Swimming Pool Clear?
Room Handover & Mess Dues Clear?
No
Yes
No
No
Yes
Mark REJECTED + Log Overdue Books/Fine
Sign Digital Signature: SPORTS
Mark REJECTED + Log Equipment Penalty
Sign Digital Signature: HOSTEL
Mark REJECTED + Log Mess Liability
Mark REJECTED + Log Lab Hardware Liability
Sign Digital Signature: Labs
Gate 1 & Gate 2 Both APPROVED?
No
Yes
Hold Accounts in MUTING_PREREQUISITE
Gate 3: Terminal Accounts & Caution Money
Unlock Gate 3: Accounts & Finance Branch
Semester Tuition Clear?
No
Yes
Mark REJECTED + Log Tuition Balance
Sign Digital Signature: ACCOUNTS + Authorize Caution Money Refund
Pipeline Blocked
Mint Official Certificate ID: PEC-CLR-YYYY-XXXXX
Compute Cryptographic SHA-256 Digest
Generate Downloadable PDF with IDA & DSA Seals
Student Clearance Completed Successfully

4. Interaction Models: Sequence Diagrams

4.1 Sequence Diagram 1: Multi-Gate Approval & Cryptographic Minting



4.2 Sequence Diagram 2: Grievance Lifecycle & Automated DAA Escalation



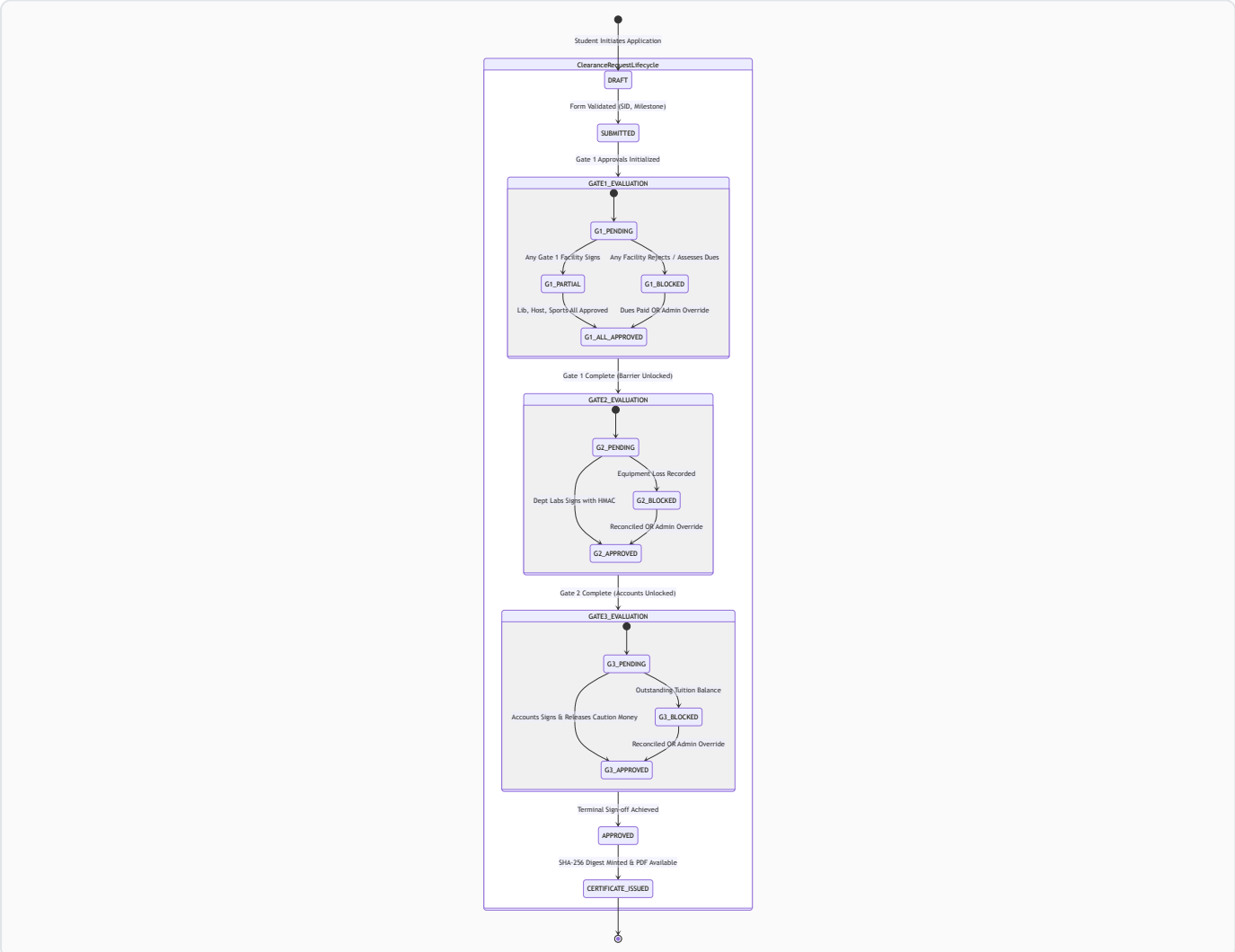
4.3 Sequence Diagram 3: Administrative Override & Concurrency Race Prevention



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5. Behavioral State Machine Diagram

This state machine models the discrete status lifecycles of both the master clearance request and individual department approval gates:



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6. System Architecture & Component Deployment Diagram

The component diagram defines the physical and logical boundaries of the system, verifying complete separation between presentation, application services, and persistent storage:

